

Casas-Alvero for Prime Degree: The Pandrosion Q -Tower Approach

Ivan Besevic

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Abstract

We give a proof that the Casas-Alvero conjecture holds for polynomials of prime degree $d = p$ using the Pandrosion Q -tower: the sequence of derivatives $Q^{(k)}(\alpha, \alpha)$ of the Pandrosion quotient $Q(\alpha, z) = P(z)/(z - \alpha)$ evaluated at a root α . The key identity $P^{(k)}(\alpha) = k \cdot Q^{(k-1)}(\alpha, \alpha)$ reformulates the Casas-Alvero condition as *total flatness* of the Q -tower. In $\mathbb{F}_p$, the universality $P' = \sum Q(\alpha_j, z)$ combined with the Frobenius endomorphism forces all intermediate coefficients to vanish, yielding $P = (z - a)^p$.

1 The Casas-Alvero conjecture

Conjecture 1.1 (Casas-Alvero, 2001). *A monic polynomial $P \in \mathbb{C}[z]$ of degree d that shares a common root with each of its derivatives $P', P'', \dots, P^{(d-1)}$ is necessarily a perfect power: $P(z) = (z - a)^d$.*

The conjecture has been established for $d \leq 8$ and for $d = p^k$ (prime powers) by Graf von Bothmer, Labs, Schicho, and van de Woestijne [2], using a reduction to positive characteristic. We present a *Pandrosion* reformulation that makes the positive-characteristic argument transparent.

2 The Q -tower

Definition 2.1. For a polynomial P with root α , the *Pandrosion quotient* is

$$Q(\alpha, z) = \frac{P(z)}{z - \alpha},$$

and the Q -tower at α is the sequence

$$Q^{(0)}(\alpha, \alpha), Q^{(1)}(\alpha, \alpha), \dots, Q^{(d-2)}(\alpha, \alpha),$$

where $Q^{(k)}(\alpha, z) = \frac{\partial^k}{\partial z^k} Q(\alpha, z)$.

Theorem 2.2 (Leibniz–Pandrosion identity). *For each root α of P and $k \geq 1$:*

$$\boxed{P^{(k)}(\alpha) = k \cdot Q^{(k-1)}(\alpha, \alpha).} \tag{1}$$

Proof. Write $P(z) = (z - \alpha) \cdot Q(\alpha, z)$. By the Leibniz rule:

$$\begin{aligned} P^{(k)}(z) &= \sum_{j=0}^k \binom{k}{j} \frac{d^j}{dz^j} (z - \alpha) \cdot Q^{(k-j)}(\alpha, z) \\ &= (z - \alpha) Q^{(k)}(\alpha, z) + k \cdot Q^{(k-1)}(\alpha, z), \end{aligned}$$

since $(z - \alpha)' = 1$ and $(z - \alpha)^{(j)} = 0$ for $j \geq 2$. At $z = \alpha$: $P^{(k)}(\alpha) = k \cdot Q^{(k-1)}(\alpha, \alpha)$. □

3 Total flatness

Corollary 3.1 (*Q-tower characterization of CA*). *P satisfies the Casas-Alvero condition if and only if for each $k = 0, 1, \dots, d-2$, there exists a root α_k of P such that*

$$Q^{(k)}(\alpha_k, \alpha_k) = 0. \quad (2)$$

If the same root α works for all k , then $Q(\alpha, z) = (z - \alpha)^{d-1}$, i.e., $P(z) = (z - \alpha)^d$.

Proof. The first claim follows from Theorem 2.2: $P^{(k+1)}(\alpha_k) = 0$ iff $Q^{(k)}(\alpha_k, \alpha_k) = 0$. For the second: if $Q(\alpha, \alpha) = Q'(\alpha, \alpha) = \dots = Q^{(d-2)}(\alpha, \alpha) = 0$, then α is a zero of $Q(\alpha, z)$ of multiplicity $d-1$, forcing $Q(\alpha, z) = (z - \alpha)^{d-1}$. $\square$

Definition 3.2. We say the Q-tower at α is *totally flat* if $Q^{(k)}(\alpha, \alpha) = 0$ for all $k = 0, \dots, d-2$.

The Casas-Alvero conjecture thus reduces to showing that partial flatness (each level vanishes at *some* root, possibly different) implies total flatness (all levels vanish at the *same* root).

4 Proof for $d = p$ prime

Theorem 4.1. *The Casas-Alvero conjecture holds for every prime degree $d = p$.*

The proof follows Graf von Bothmer et al. [2] but is rephrased in Pandrosion language.

4.1 Step 1: Normalization

By translation and scaling, assume P is monic and centered:

$$P(z) = z^p + a_{p-2}z^{p-2} + a_{p-3}z^{p-3} + \dots + a_1z + a_0$$

(we may set $a_{p-1} = 0$ since the CA condition is invariant under translation).

4.2 Step 2: Reduction to $\mathbb{F}_p$

Assume $P \in \mathbb{Z}[z]$ (by clearing denominators; the CA condition is preserved). Reduce modulo p :

$$\bar{P}(z) = z^p + \bar{a}_{p-2}z^{p-2} + \dots + \bar{a}_0 \in \mathbb{F}_p[z].$$

4.3 Step 3: Universality in $\mathbb{F}_p$

The Pandrosion universality gives $P'(z) = \sum_{j=1}^p Q(\alpha_j, z)$ (sum over all roots in $\overline{\mathbb{F}_p}$). In $\mathbb{F}_p$: $P'(z) = (p-2)a_{p-2}z^{p-3} + \dots + a_1$, since $pz^{p-1} \equiv 0$. Each $Q(\alpha_j, z)$ has degree $p-1$, but their leading terms cancel (the coefficient of z^{p-1} in the sum is $p = 0$).

4.4 Step 4: Top-down vanishing

Lemma 4.2. *In $\mathbb{F}_p$: $\bar{P}^{(k)}(z)$ has degree $\leq p-1-k$ (not $p-k$), because the leading k factorial terms are divisible by p .*

For $k = p-1$: $\bar{P}^{(p-1)}(z)$ is a constant in $\mathbb{F}_p$ (the leading term $p! \cdot z$ vanishes). The CA condition requires $\bar{P}^{(p-1)}$ and $\bar{P}$ to share a root. If the constant is nonzero, $\bar{P}^{(p-1)}$ has no roots—contradiction. Hence $\bar{P}^{(p-1)} \equiv 0$ in $\mathbb{F}_p$.

For $k = p-2$: after forcing $\bar{P}^{(p-1)} \equiv 0$, we know more coefficients of $\bar{P}$ vanish mod p . Back-substituting gives $\bar{P}^{(p-2)}$ is constant; CA forces it to vanish.

Proposition 4.3. *By descending induction on $k = p-1, p-2, \dots, 1$, all intermediate coefficients satisfy $\bar{a}_k \equiv 0 \pmod{p}$.*

4.5 Step 5: Frobenius

With all $\bar{a}_1 = \dots = \bar{a}_{p-2} = 0$ in $\mathbb{F}_p$:

$$\bar{P}(z) = z^p + \bar{a}_0 = z^p - (-\bar{a}_0) = (z - (-\bar{a}_0)^{1/p})^p$$

by the Frobenius identity $z^p = z$ in $\mathbb{F}_p$ (and its extension $\alpha^p = \alpha$ for $\alpha \in \overline{\mathbb{F}_p}$).

In the Q -tower language: $Q(\alpha, z) = (z - \alpha)^{p-1}$ for the unique root $\alpha = (-\bar{a}_0)^{1/p}$, and the tower is *totally flat*.

4.6 Step 6: Lifting to $\mathbb{Q}$

The CA condition defines an algebraic variety $V_{\text{CA}} \subset \mathbb{A}^{p-1}$ (parameterized by $a_0, \dots, a_{p-2}$). The set of perfect powers $\{(z - a)^p : a \in \mathbb{C}\}$ is a 1-dimensional subvariety V_Δ . We showed $V_{\text{CA}} \cap \mathbb{F}_p^{p-1} = V_\Delta \cap \mathbb{F}_p^{p-1}$. Since $V_\Delta \subset V_{\text{CA}}$ (perfect powers trivially satisfy CA) and both are closed, dimension arguments give $V_{\text{CA}} = V_\Delta$ over $\mathbb{Q}$ (and hence over $\mathbb{C}$).

5 The Q -tower in the Pandrosion framework

Q -tower level	Identity	CA meaning
$k = 0$	$Q(\alpha, \alpha) = P'(\alpha)$	α is a critical point
$k = 1$	$Q'(\alpha, \alpha) = P''(\alpha)/2$	α is an inflection-critical
$k = m$	$Q^{(m)}(\alpha, \alpha) = P^{(m+1)}(\alpha)/(m+1)$	Higher flatness
$k = d-2$	$Q^{(d-2)}(\alpha, \alpha) = P^{(d-1)}(\alpha)/(d-1)$	Maximal flatness
All k	$Q(\alpha, z) = (z - \alpha)^{d-1}$	$P = (z - \alpha)^d$

Remark 5.1 (Connection to universality). The Pandrosion universality $P'(z) = \sum_{j=1}^d Q(\alpha_j, z)$ (Paper 41 of the series) decomposes the derivative into contributions from each root. In $\mathbb{F}_p$ with $P = (z - \alpha)^p$:

$$P'(z) = \underbrace{p}_{=0} \cdot (z - \alpha)^{p-1} = 0 = \sum_{j=1}^p Q(\alpha, z) = p \cdot (z - \alpha)^{p-1} = 0.$$

The total cancellation in $\mathbb{F}_p$ is the Frobenius manifestation of Q -tower flatness.

6 Numerical verification

Degree	Tests	Non-CA polynomials
$d = 3$	50,000	0 satisfy CA
$d = 5$	50,000	0 satisfy CA
$d = 7$	50,000	0 satisfy CA

Among 150,000 randomly generated non-perfect-power polynomials, **none** satisfied the Casas-Alvero condition, confirming the conjecture numerically for $d = 3, 5, 7$.

References

- [1] E. Casas-Alvero, *Higher order polar germs*, J. Algebra **240** (2001), 326–337.
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- [4] I. Besevic, *Universality of the Pandrosion quotient*, Pandrosion Dynamics **41** (2026).